



DOC#:01.13.00LidarClient02

# **YDLIDAR LIDARCLIENT AND ROS USER MANUAL**

[www.ydlidar.com](http://www.ydlidar.com)

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## 1 OPERATION UNDER WINDOWS

### 1.1 Device Connection

When YDLIDAR LiDAR is evaluated and developed under windows, YDLIDAR LiDAR and PC need to be interconnected.

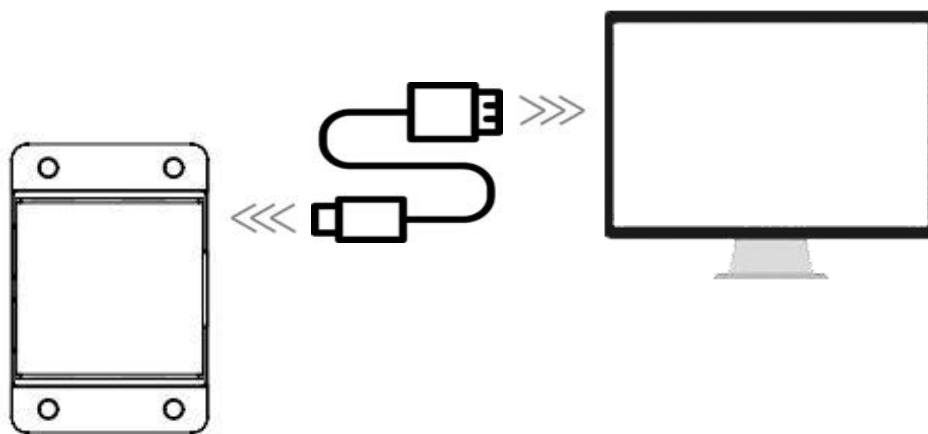


FIG 1 YDLIDAR LIDAR DEVICE CONNECTION

Connect the adapter board with YDLIDAR LiDAR first, then connect the USB cable to the USB port of the adapter board and the PC. Note that the Type-C interface of the USB cable is connected to the USB\_DATA of the USB interface board, and the idle mode is used after YDLIDAR LiDAR is powered on.

The drive current of USB interface of some development platforms or PC is not sufficient. YDLIDAR LiDAR need to be connected to the auxiliary power supply of +5V(Except for GS Serials LiDAR), otherwise the lidar will be abnormal.

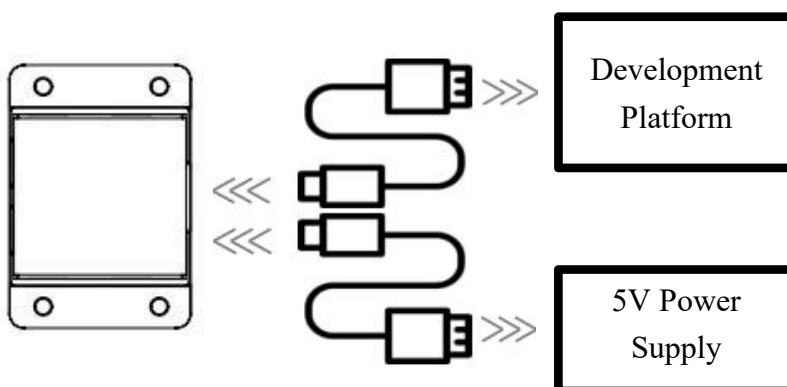


FIG 2 YDLIDAR LIDAR AUXILIARY POWER SUPPLY

### 1.2 Driver Installation

To evaluate and develop the YDLIDAR LiDAR under Windows, you need to install the serial port driver of the USB adapter board. The USB adapter board of this kit adopts

CP2102 chip to realize serial port (UART) to USB signal conversion. Its driver can be downloaded from our official website.

<https://www.ydlidar.com/download/category/lidar-sensor>

After unzipping the driver package, run the CP2102's Windows driver installation file (exe file under CP210x\_VCP\_Windows). Please select the 32-bit version (x86) or 64-bit version (x64) installation program according to the version of the windows operating system.

| Name                                   | Type              | Size     |
|----------------------------------------|-------------------|----------|
| x64                                    | File folder       |          |
| x86                                    | File folder       |          |
| CP210xVCPInstaller_x64.exe             | Application       | 1,034 KB |
| CP210xVCPInstaller_x86.exe             | Application       | 911 KB   |
| dpinst.xml                             | XML Document      | 12 KB    |
| SLAB_License_Agreement_VCP_Windows.txt | 文本文档              | 9 KB     |
| slabvcp.cat                            | Security Catalog  | 11 KB    |
| slabvcp.inf                            | Setup Information | 12 KB    |

FIG 3 YDLIDAR LIDAR DRIVER VERSION SELECTION

Double-click the exe file and follow the prompts to install it.



FIG 4 YDLIDAR LIDAR DRIVER INSTALLING

After the installation is complete, you can right-click on [My Computer] and select [Properties]. On the open [System] screen, select [Device Manager] from the left menu to access the Device Manager.

Expand [Port] to see the serial port name corresponding to the identified USB adapter, that is, the driver installation is successful. The following figure shows COM3. (Note that the port must be checked in case of YDLIDAR LiDAR and PC interconnection).

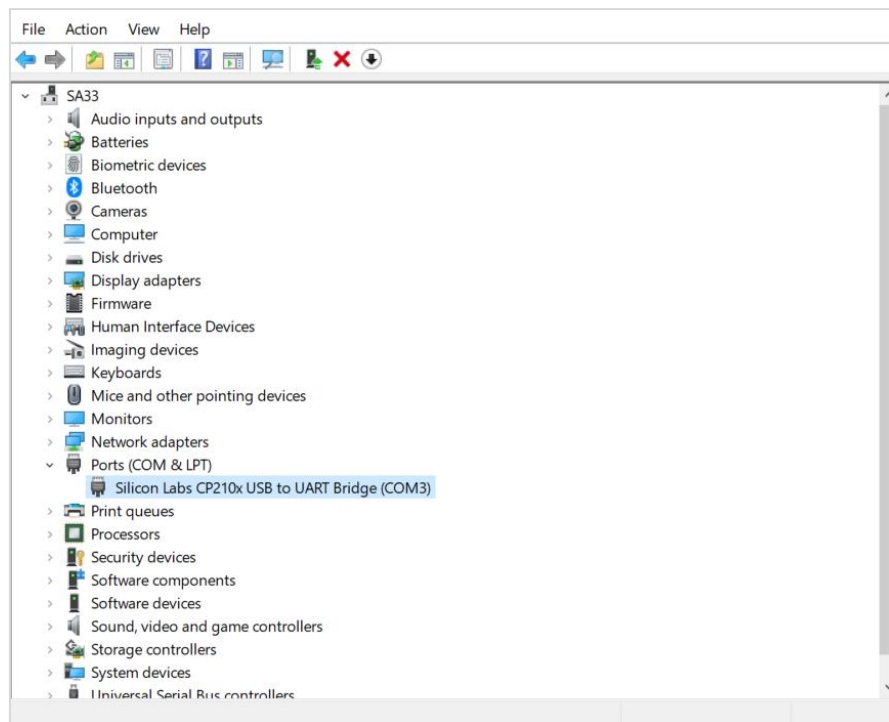


FIG 5 YDLIDAR LIDAR DRIVER INSTALLATION CHECK

## 1.3 How to Use LidarClient software

YDLIDAR provides EaiLidarTest, a point cloud data LidarClient software for YDLIDAR LiDAR real-time scanning. Users can use this software to visually observe the YDLIDAR LiDAR scanning map: YDLIDAR provides YDLIDAR LiDAR real-time point cloud data and real-time scanning frequency. At the same time, the scanned data can be saved offline to an external file for further analysis. LidarClient software download link:

<https://www.ydlidar.com/download/category/lidar-sensor>

Before using the YDLIDAR software, make sure that the YDLIDAR LiDAR USB adapter board serial port driver is installed successfully, and interconnect the YDLIDAR LiDAR with the USB port of the PC.

Run the LidarClient software: EaiLidarTest.exe, by default, it will automatically log in and enter the main interface, as below image shown:

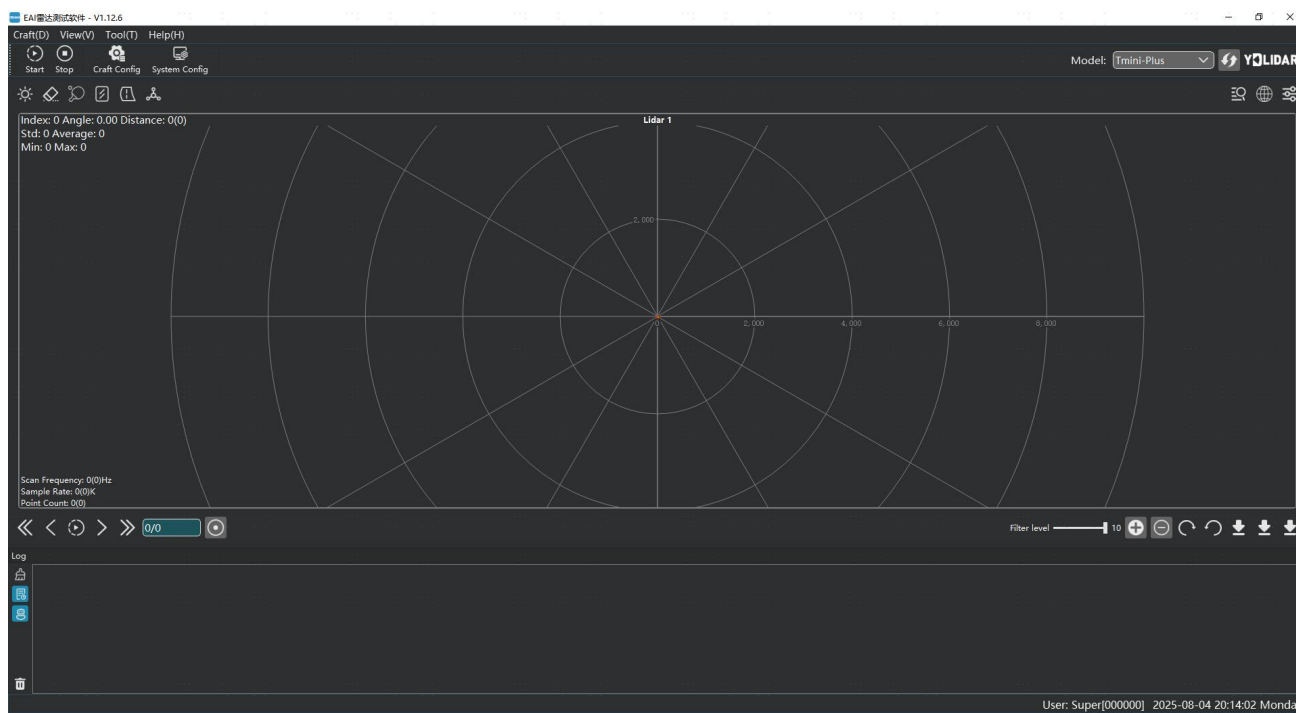


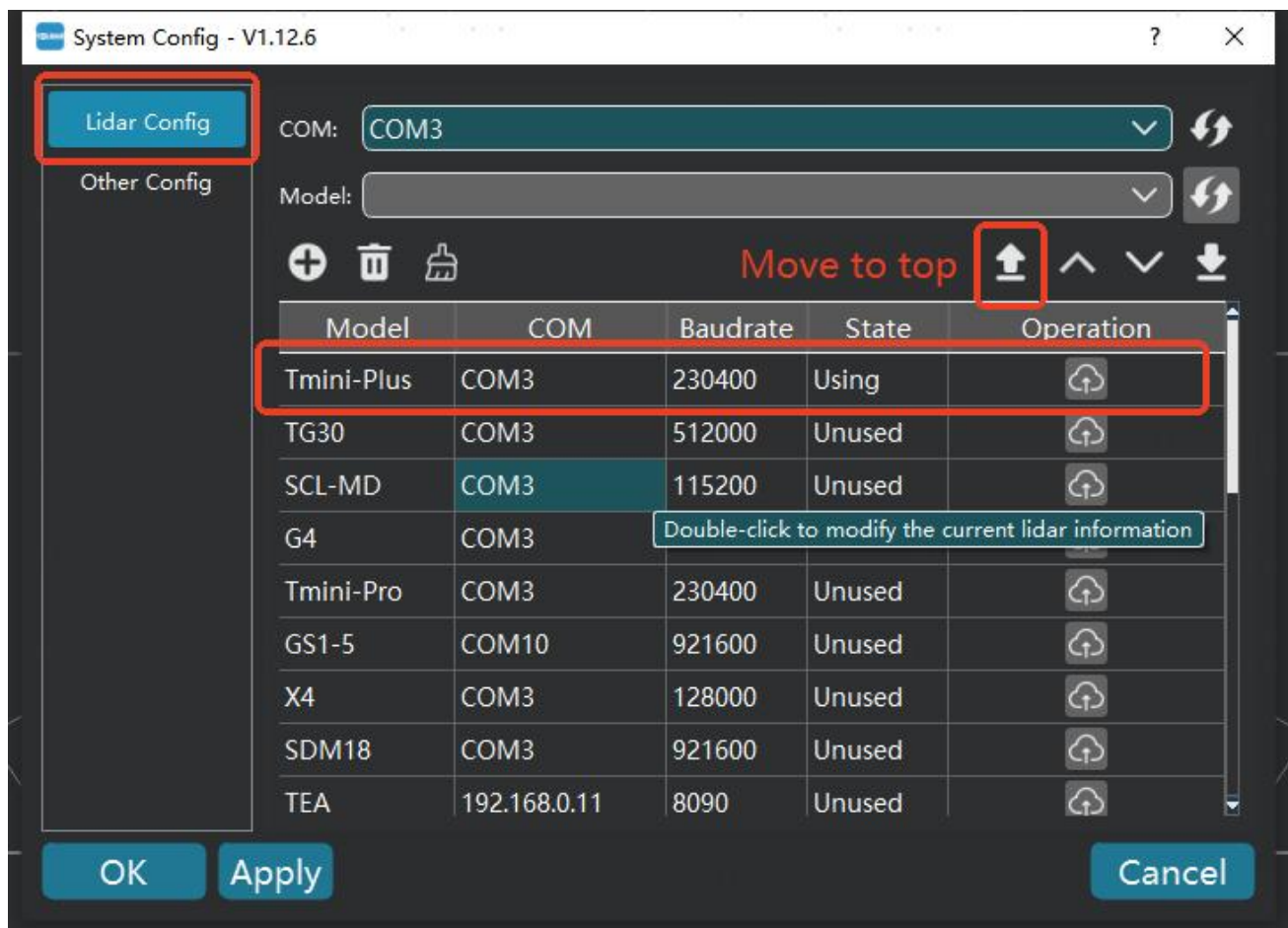
FIG 6 YDLIDAR LIDAR LIDARCLIENT SOFTWARE INTERFACE

### 1.3.1 Start Scanning

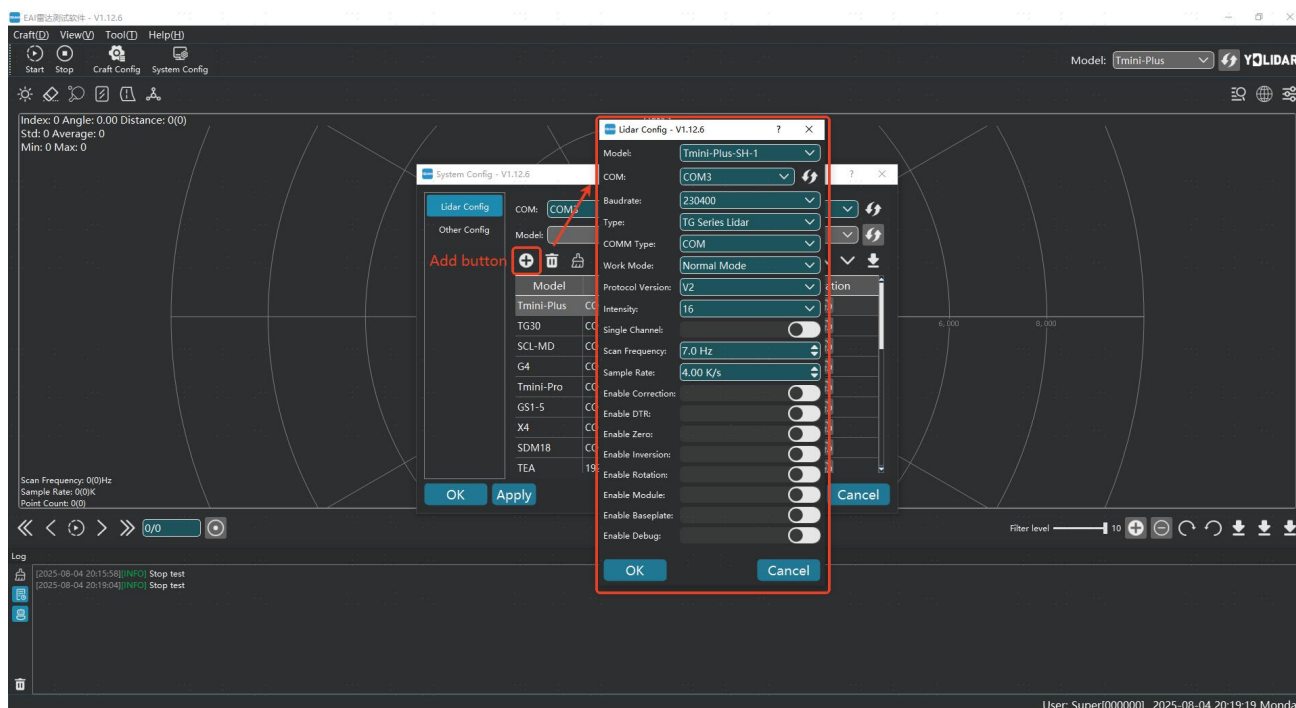
Click "System Config" button, then click "Lidar Config" to find out if there is LiDAR model you need. If so, click the "move to the top button " to the top, and select the corresponding serial port. If can not find out serial port accordingly, click the "Refresh Serial Port" button on the right and then select the serial port. After the configuration is completed, click the "OK" button in the lower right corner to save and close the system configuration window. The configuration list is shown in the figure below.

| LiDAR Option   | LiDAR Model Option               |
|----------------|----------------------------------|
| Tmini-Plus     | YDLIDAR Tmini Plus (12meter)     |
| Tmini-Plus-SH  | YDLIDAR Tmini Plus (25meter)、TW3 |
| G4             | YDLIDAR G2、G4                    |
| GS1-5          | YDLIDAR GS2、GS5                  |
| TG15/TG30/TG50 | YDLIDAR TG15/TG30/TG50           |
| SDM15/SDM18    | YDLIDAR SDM15/SDM18              |
| X2/X3/X4       | YDLIDAR X2/X3/X4                 |
| X4-Pro         | YDLIDAR X4 Pro                   |
| TEA            | YDLIDAR TEA                      |
| TIA            | YDLIDAR TIA                      |
| TIA-H          | YDLIDAR TIA-H                    |

Configuration Interface (For example:YDLIDAR Tmini Plus ) in below:



If can not find out LiDAR model you need, you need to manually add the configuration items for the model. Click "+" button and fill in the corresponding parameters in the pop-up interface, as shown below. If you do not know how to fill in the parameters, please contact YDLIDAR after-sales team for help directly by [support@ydlidar.com](mailto:support@ydlidar.com).



Click "Start" button to start scanning. If the startup is successful, the point cloud of the current LiDAR will be displayed in real time in the polar coordinate diagram on the main interface, as shown in the figure below.

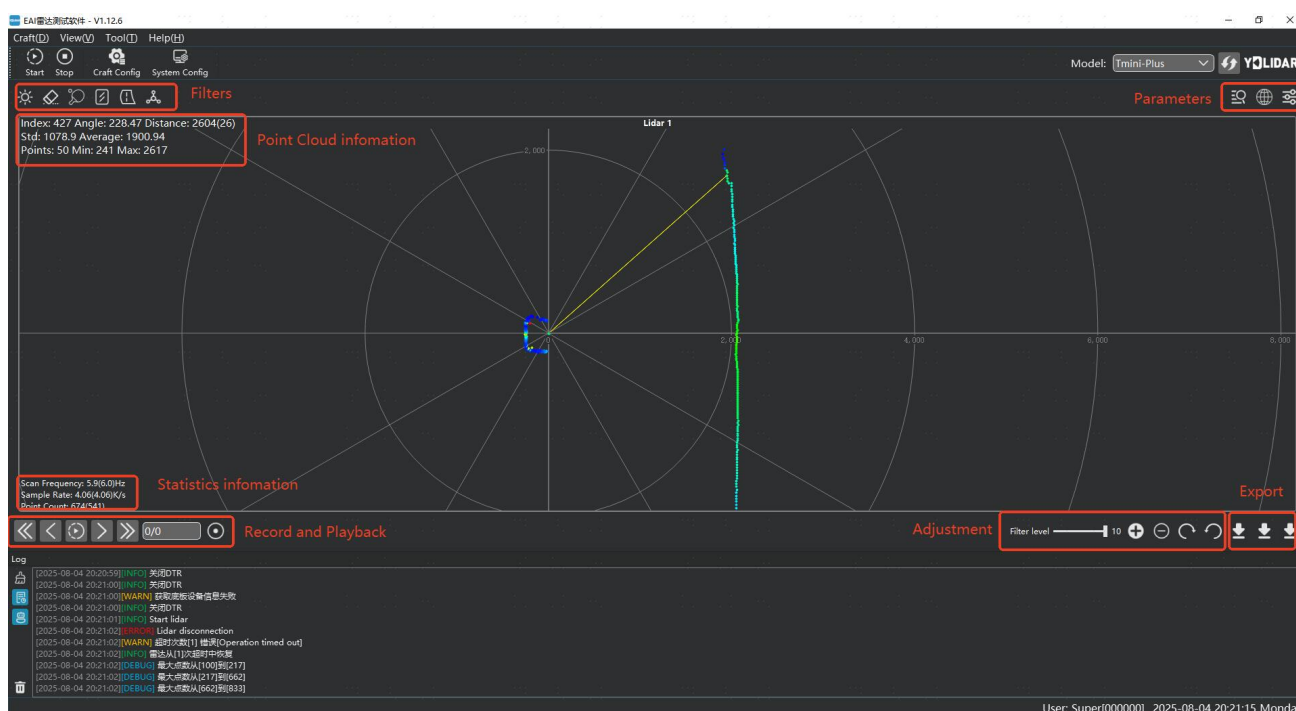


FIG 7 LIDAR SCANNING POINT CLOUD DISPLAY

## 1.3.2 Data Storage

During lidar scanning, click [File] in the main menu, select [Export to CSV], and save point cloud data according to the prompts. Then the system will save the point cloud information scanned in a circle in CSV format.

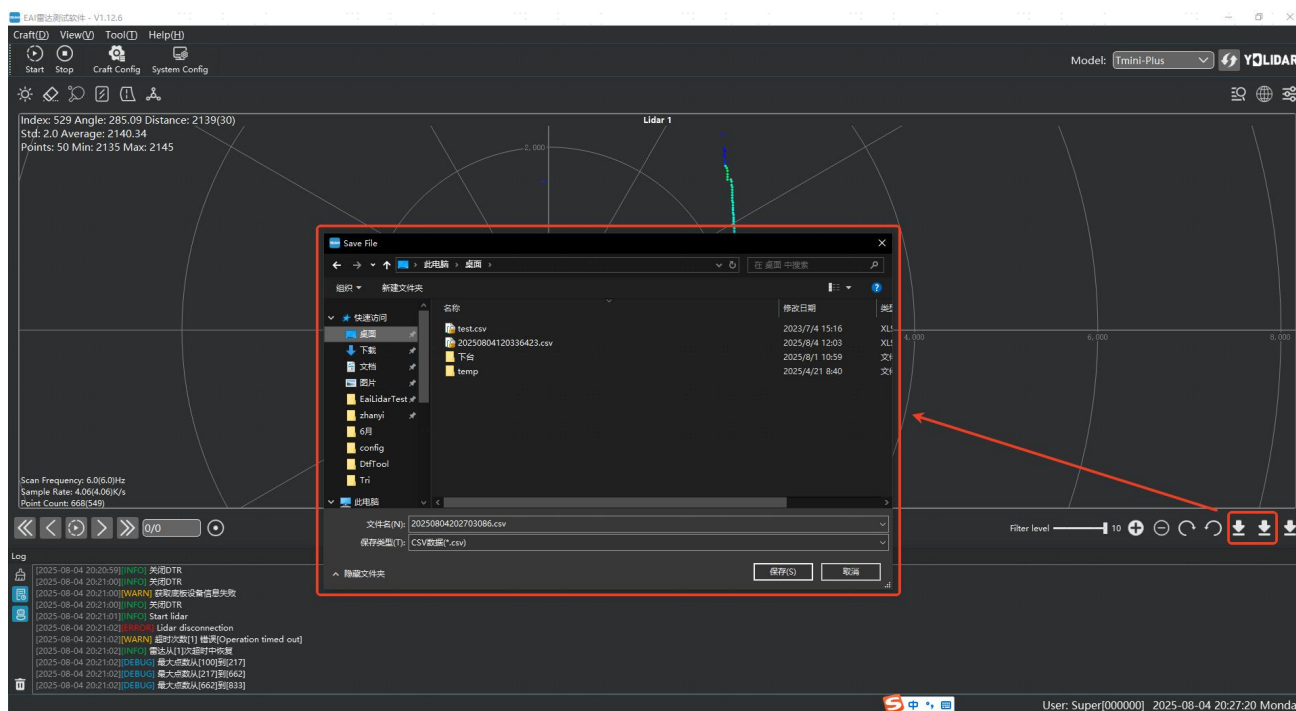


FIG 8 SAVE DATA

### 1.3.3 Show Intensity Graph

Right-click the mouse and select "Show Intensity Graph" in the pop-up menu to display the intensity values of a total of 50 points on the left and right of the mouse lock position., as shown below.

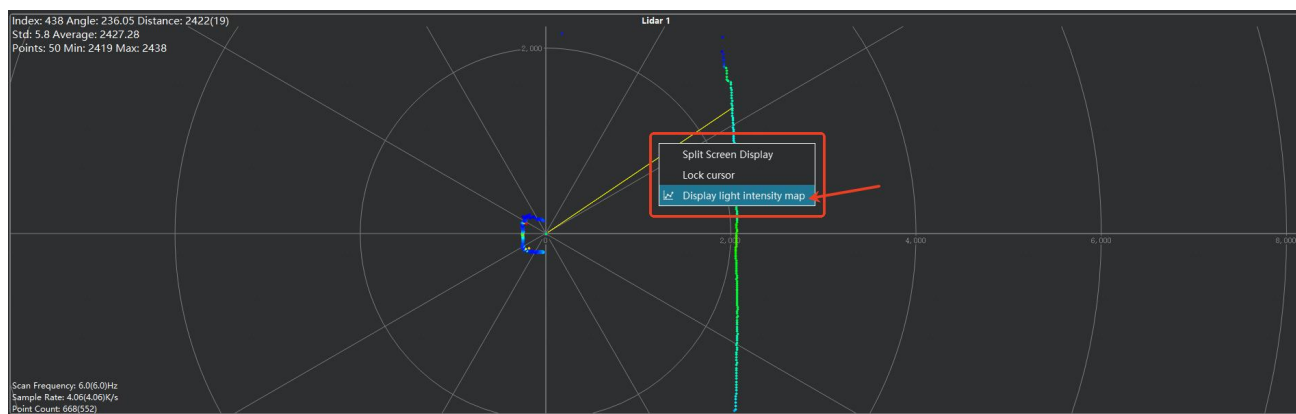


FIG 9 RIGHT-CLICK MENU

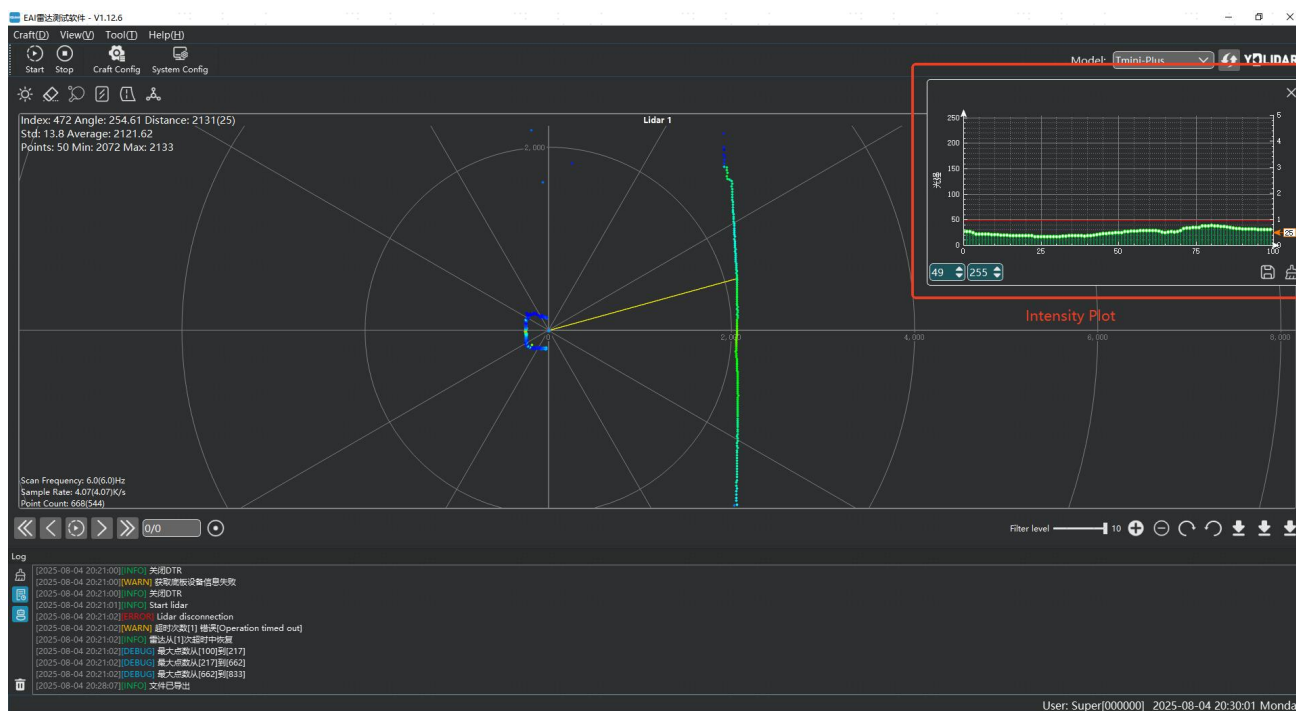
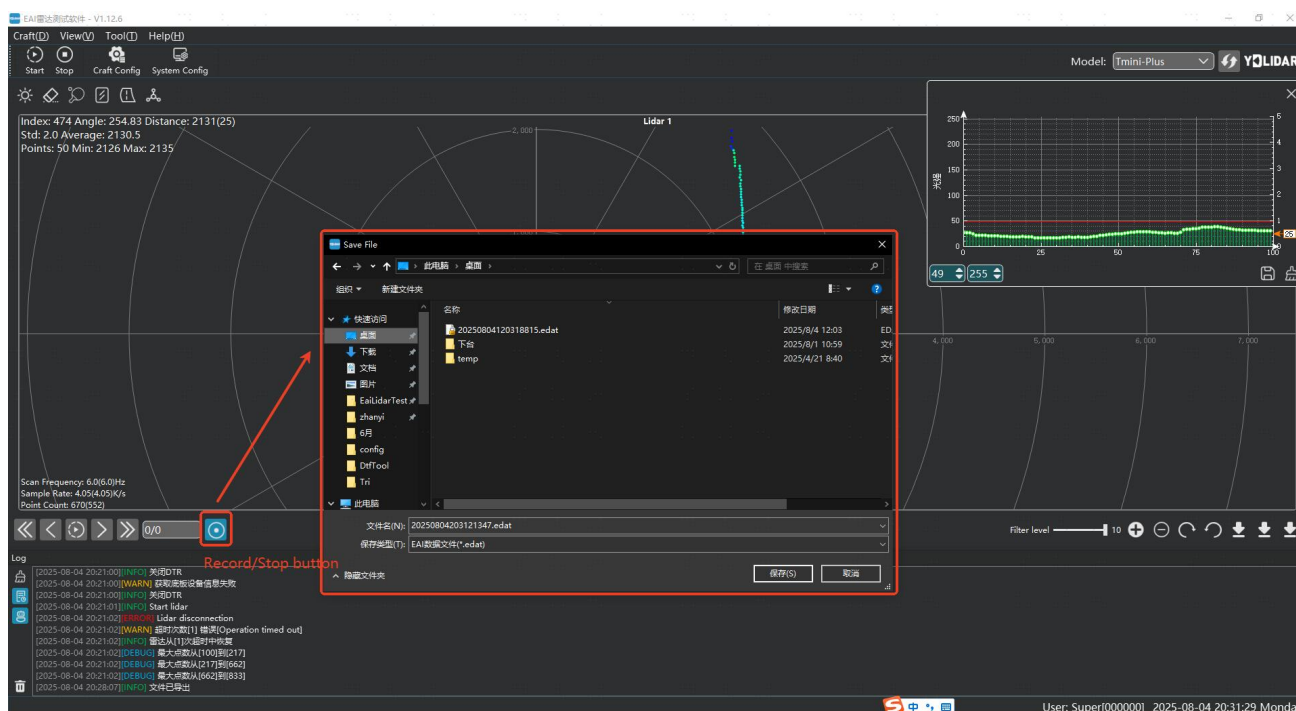


FIG 10 SHOW INTENSITY GRAPH

### 1.3.4 Play and Record

When start to scan, you can click "Record" button in the lower left corner and enter file name to start recording. If you do not need to record, click the "Record" button again to stop recording.



If you need to play the recorded point cloud, you can click the "Playback" button and select the file saved during recording to play the point cloud. You can pause the playback and manually switch frames, as shown in the figure below.

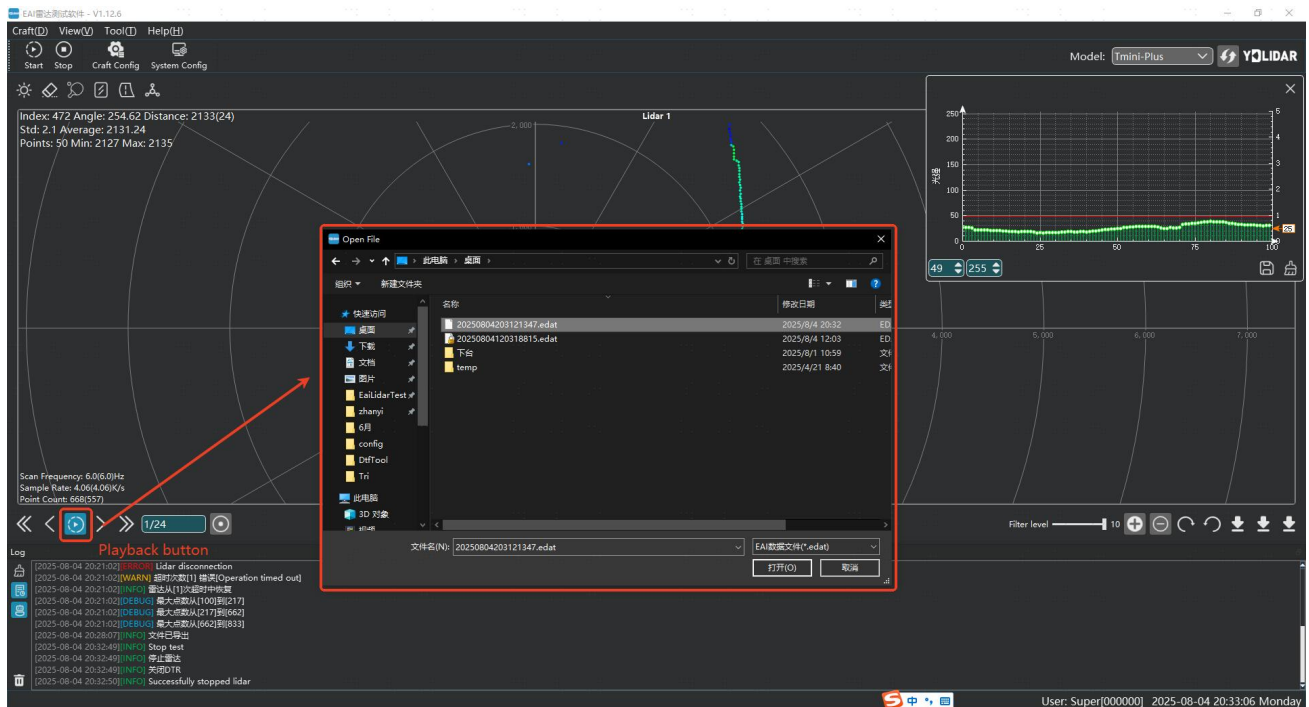


FIG 11 RECORD AND PLAY

### 1.3.5 Debug

If you fail to connect LiDAR, you need to enable debugging to check the LiDAR's raw data and analyze it in conjunction with the log in the log bar. You can click [System Configuration -> LiDAR Configuration -> Double-click the LiDAR item -> Click "Enable Debugging" in the pop-up interface, as shown in the figure below (For example: T-mini Plus).

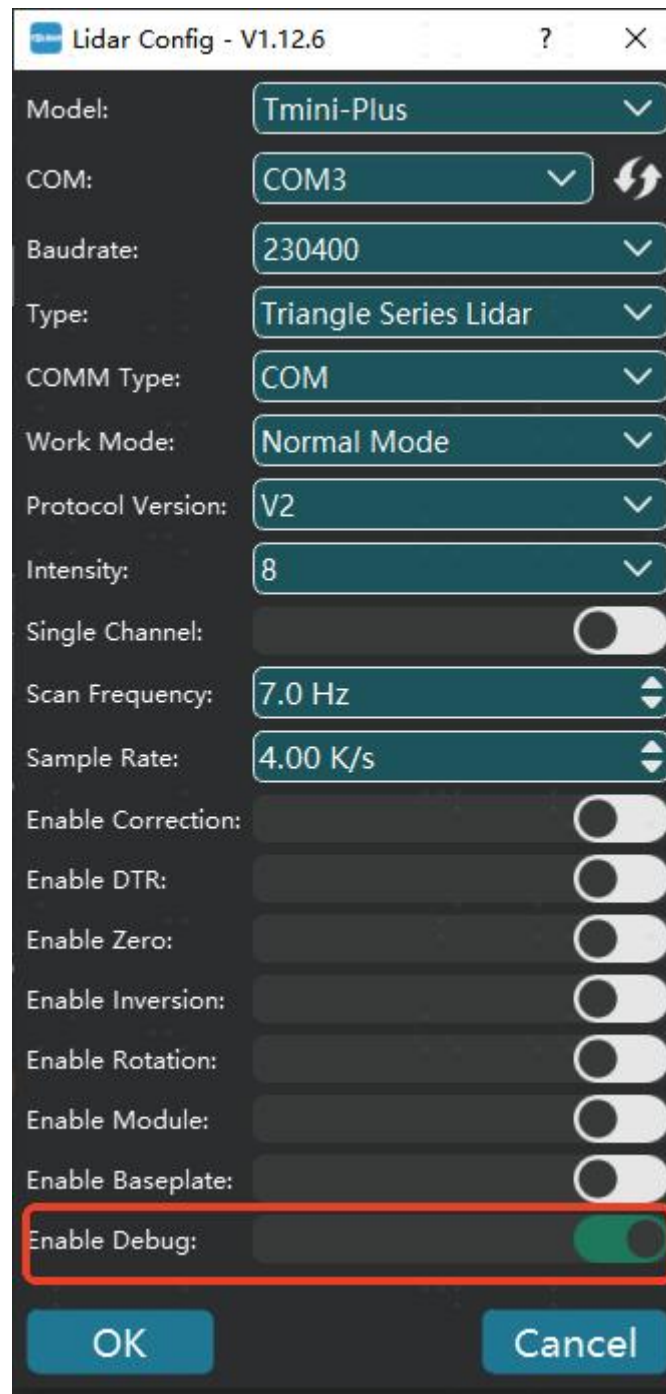


FIG 12 START DEBUGGING

### 1.3.6 Filter

Please do not use filter function if unnecessary

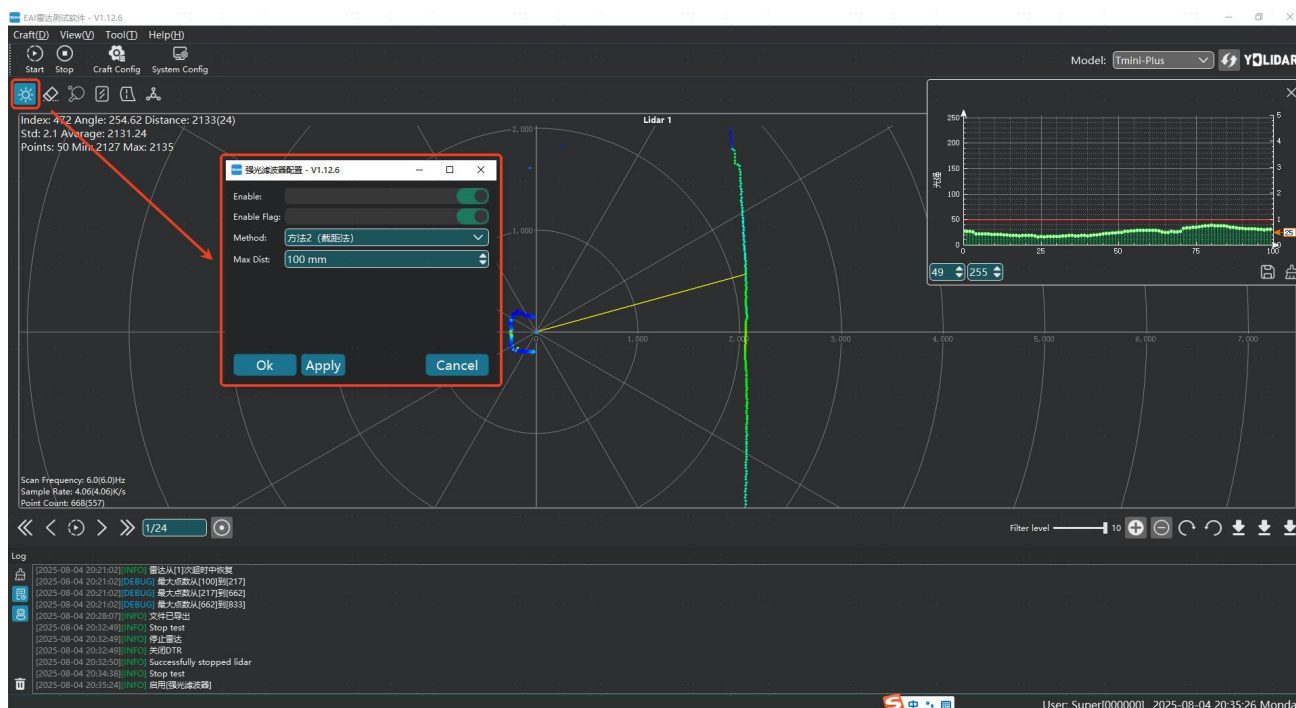


FIG 13 FILTER SETTING

Note: For any issues about LiDARClient software, please contact for help through YDLIDAR after-sales team [support@ydlidar.com](mailto:support@ydlidar.com) directly.

## 2 LINUX ROS OPERATION

There are many Linux versions, this article only uses Ubuntu 18.04, Melodic version ROS as an example.

SDK driver address:

<https://github.com/YDLIDAR/YDLidar-SDK>

ROS driver address:

[https://github.com/YDLIDAR/ydlidar\\_ros\\_driver](https://github.com/YDLIDAR/ydlidar_ros_driver)

ROS2 driver address:

[https://github.com/YDLIDAR/ydlidar\\_ros2\\_driver/tree/humble](https://github.com/YDLIDAR/ydlidar_ros2_driver/tree/humble)

### 2.1 Device Connection

Under Linux, the YDLIDAR LiDAR and PC interconnect processes are consistent with those under Windows. See [Device Connection under Window](#).

## 2.2 Compile and Install YDLidar-SDK

ydliar\_ros\_driver depends on the YDLidar-SDK library. If you have never installed the YDLidar-SDK library, or it has expired, you must first install the YDLidar-SDK library. If you have the latest version of YDLidar-SDK installed, please skip this step, then go to the next step.

```
$ git clone https://github.com/YDLIDAR/YDLidar-SDK.git
$ cd YDLidar-SDK/build
$ cmake ..
$ make
$ sudo make install
```

## 2.3 ROS Driver Installation

1) Cloning GitHub's ydliar\_ros\_driver Package:

```
$ git clone https://github.com/YDLIDAR/ydliar_ros_driver.git
ydliar_ws/src/ydliar_ros_driver
```

2) Build the ydliar\_ros\_driver software package:

```
$ cd ydliar_ws
$ catkin_make
```

3) Package environment Settings:

```
$ source ./devel/setup.sh
```

Note: Add a permanent workspace environment variable. It will be very convenient if ROS environment variables are automatically added to your bash session every time you start a new shell:

```
$ echo "source ~/ydliar_ws/devel/setup.bash" >> ~/.bashrc
$ source ~/.bashrc
```

4) Verify that your package path is set, echo the ROS\_PACKAGE\_PATH variable.

```
$ echo $ROS_PACKAGE_PATH
```

You should see something like this: /home/tony/ydliar\_ws/src:/opt/ros/melodic/share.

## 5) Create Serial Port Alias [Optional]

```
$ chmod 0777 src/ydlidar_ros_driver/startup/*
$ sudo sh src/ydlidar_ros_driver/startup/initenv.sh
```

Note: After completing the previous operation, re-insert the LiDAR again.

## 2.4 Run the ydlidar\_ros\_driver

Run ydlidar\_ros\_driver with startup file, Example (Tmini Plus) is shown below:

```
$ roslaunch ydlidar_ros_driver Tmini.launch
```

## 2.5 RVIZ Scan Result Checking

```
$ roslaunch ydlidar_ros_driver lidar_view.launch
```

Note: Take G4 as an example by default. If you use other lidars, change the lidar.launch file in lidar\_view.launch file to the corresponding \*\*. (If using YDLIDAR T-mini Plus lidar, change to Tmini.launch)

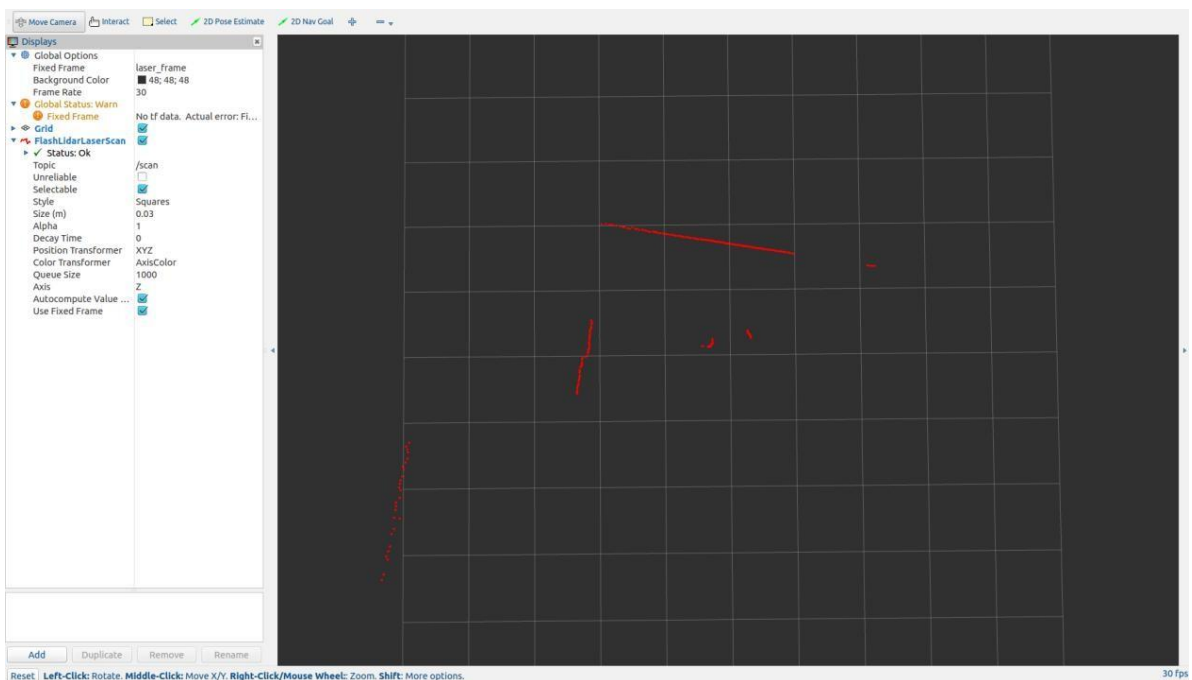


FIG 14 YDLIDAR T-MINI PLUS RVIZ

## 2.6 Modify Scan Angle

The scanning data seen by running the launch file is displayed by default with 360-degree data. To modify the display range, you need to modify the configuration parameters in the launch file. The specific operation is as follows:

- 1) Switch to the directory where the corresponding [launch file] is located, edit the file, and its content is as shown in the figure:

```
$ vim Tmini.launch
```

```
<launch>
  <node name="ydlidar_lidar_publisher" pkg="ydlidar_ros_driver" type="ydlidar_ros_driver_node" output="screen" respawn="false" >
    <!-- string property -->
    <param name="port" type="string" value="/dev/ydlidar"/>
    <param name="frame_id" type="string" value="laser_frame"/>
    <param name="ignore_array" type="string" value=""/>

    <!-- int property -->
    <param name="baudrate" type="int" value="230400"/>
    <!-- 0:TYPE_TOF, 1:TYPE_TRIANGLE, 2:TYPE_TOF_NET -->
    <param name="lidar_type" type="int" value="1"/>
    <!-- 0:YDLIDAR_TYPE_SERIAL, 1:YDLIDAR_TYPE_TCP -->
    <param name="device_type" type="int" value="0"/>
    <param name="sample_rate" type="int" value="4"/>
    <param name="abnormal_check_count" type="int" value="4"/>

    <!-- bool property -->
    <param name="resolution_fixed" type="bool" value="false"/>
    <param name="auto_reconnect" type="bool" value="true"/>
    <param name="reversion" type="bool" value="true"/>
    <param name="inverted" type="bool" value="true"/>
    <param name="isSingleChannel" type="bool" value="false"/>
    <param name="intensity" type="bool" value="true"/>
    <param name="support_motor_dtr" type="bool" value="false"/>
    <param name="invalid_range_is_inf" type="bool" value="false"/>
    <param name="point_cloud_preservative" type="bool" value="false"/>

    <!-- float property -->
    <param name="angle_min" type="double" value="-180" />
    <param name="angle_max" type="double" value="180" />
    <param name="range_min" type="double" value="0.0" />
    <param name="range_max" type="double" value="16.0" />
    <param name="frequency" type="double" value="6.0"/>
  </node>
  <node pkg="tf" type="static_transform_publisher" name="base_link_to_laser4"
    args="0.0 0.0 0.2 0.0 0.0 0.0 /base_footprint /laser_frame 40" />
</launch>
```

FIG 15 TMINI.LAUNCH FILE

Note: For more information about the file contents, please refer to

[https://github.com/YDLIDAR/ydlidar\\_ros\\_driver#configure-ydlidar\\_ros\\_driver-internal-parameter](https://github.com/YDLIDAR/ydlidar_ros_driver#configure-ydlidar_ros_driver-internal-parameter)

- 2) The YDLIDAR T-mini Plus lidar coordinates follow the right-hand rule within ROS, with an angle range of [-180, 180]. "angle\_min" is the start angle, and "angle\_max" is the endangle. The specific scope needs to be modified according to actual use.

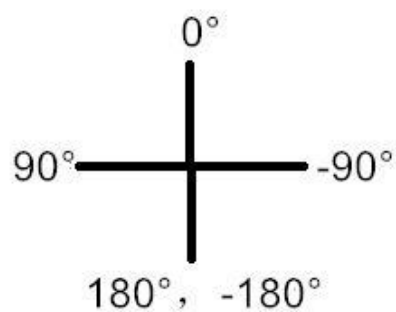


FIG 16 YDLIDAR T-MINI PLUS COORDINATES DEFINITION

### 3 Revise

| Date       | Version | Content          |
|------------|---------|------------------|
| 2026-03-30 | 1.0     | The 1st released |